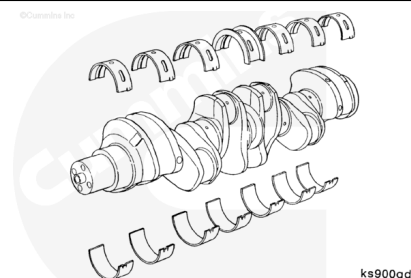


Trainings ▾

General Information

The crankshaft uses forged counterweights.

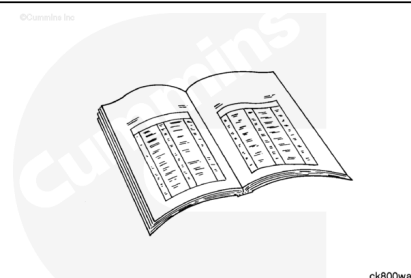
Oversize main bearings, thrust bearings, and connecting rod bearings are available for service. Cummins Inc. recommends regrounding all the main or connecting rod journals when one requires regrounding. See the appropriate parts catalog.



Preparatory Steps

⚠ WARNING ⚠

The engine lifting equipment must be designed to lift the engine and transmission as an assembly without causing personal injury.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with the skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged

contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

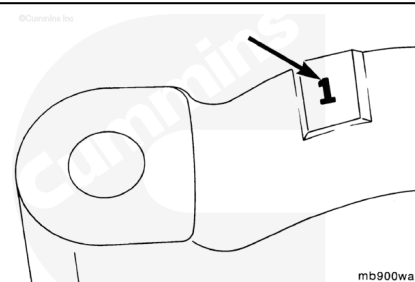
- Disconnect the batteries. Refer to the original equipment manufacturer (OEM) service manual.
- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/216/216-007-037.html>)
- Remove the engine and place it on an engine stand. Refer to Procedure 000-001 in Section 0. (</qs3/pubsys2/xml/en/procedures/216/216-000-001.html>)
- Remove the drive belt. Refer to Procedure 008-002 in Section 8. (</qs3/pubsys2/xml/en/procedures/220/220-008-002-tr.html>)
- Remove the vibration damper, if installed. Refer to Procedure 001-052 in Section 1. (</qs3/pubsys2/xml/en/procedures/216/216-001-052-tr.html>)
- Remove the front cover. Refer to Procedure 001-031 in Section 1. (</qs3/pubsys2/xml/en/procedures/216/216-001-031-tr.html>)
- Remove the starter. Refer to Procedure 013-020 in Section 13. (</qs3/pubsys2/xml/en/procedures/216/216-013-020-tr.html>)
- Remove the flexplate, if installed. Refer to Procedure 016-004 in Section 16. (</qs3/pubsys2/xml/en/procedures/216/216-016-004-tr.html>)
- Remove the flywheel, if installed. Refer to Procedure 016-005 in Section 16. (</qs3/pubsys2/xml/en/procedures/216/216-016-005-tr.html>)
- Remove the flywheel housing. Refer to Procedure 016-006 in Section 16. (</qs3/pubsys2/xml/en/procedures/216/216-016-006-tr.html>)

- For front gear train engines, remove the rear seal and rear seal carrier. Refer to Procedure 001-024 in Section 1. (</qs3/pubsys2/xml/en/procedures/216/216-001-024-tr.html>)
- Remove the lubricating oil suction tube. Refer to Procedure 007-035 in Section 7. (</qs3/pubsys2/xml/en/procedures/216/216-007-035-tr.html>)
- Remove the block stiffener plate. Refer to Procedure 001-089 in Section 1. (</qs3/pubsys2/xml/en/procedures/220/220-001-089.html>)
- Remove the fuel pump and gear. Refer to Procedure 005-016 in Section 5. (</qs3/pubsys2/xml/en/procedures/216/216-005-016-tr.html>)
- Remove the camshaft gear. Refer to Procedure 001-012 in Section 1. (</qs3/pubsys2/xml/en/procedures/216/216-001-012-tr.html>)
- Remove the rear gear housing. 001-034 in Section 1.
- Remove the connecting rod caps. Refer to Procedure 001-005 in Section 1. (</qs3/pubsys2/xml/en/procedures/220/220-001-005.html>)

Remove

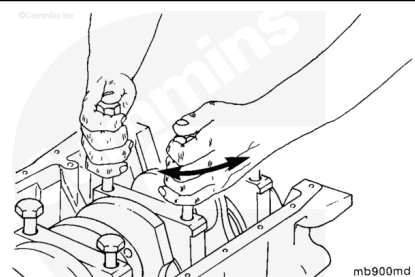
Before removing the main bearing caps, make certain that the caps are clearly marked for their location on the lubricating oil cooler side of the main bearing cap and cylinder block.

The number 1 cap is at the front of the engine.



⚠ CAUTION ⚠

Do not pry on the main bearing caps to free them from the cylinder block. Damage to the main bearing caps and cylinder block can result.



Loosen the main bearing capscrews completely, but do **not** remove.

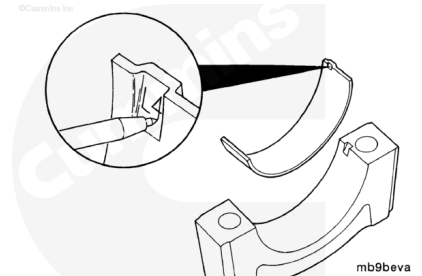
Use two of the main bearing cap bolts to “wiggle” the main bearing cap loose, being careful **not** to damage the bolt threads.

Remove the main bearing cap.

Mark the lower main bearings for position and number as they are removed.

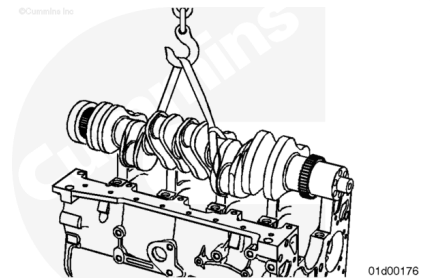
Use an awl to mark the bearing position in the tang area.

Note : Mark the bearing's position for future identification or possible failure analysis.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Note : Lift the crankshaft straight up to avoid damage to the crankshaft and cylinder block.

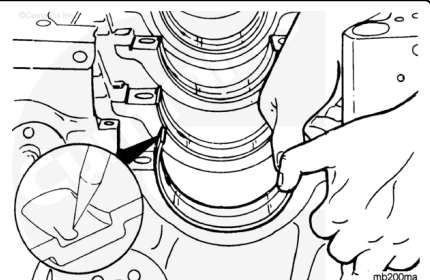
Install nylon lift sling, Part Number 3375957, around the number 3 and number 4 rod bearing journals.

Attach the sling to a hoist and remove the crankshaft.

Remove the upper main bearings.

Use an awl to mark the bearing's position in the tang area.

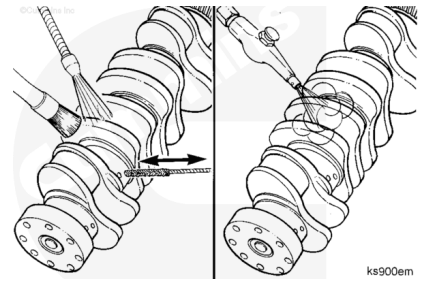
Note : Mark the bearing's position for future identification or possible failure analysis.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Steam clean or use hot, soapy water to clean the crankshaft and gear(s).

Dry with compressed air. Make sure to blow out the threaded holes on each end of the crankshaft and the oil drillings.

Use a bristle brush to clean the oil drillings.

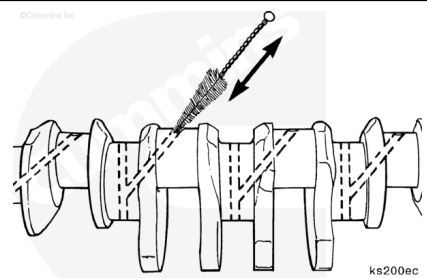
Use a fine crocus cloth to polish the machined surfaces.

Inspect the front crankshaft gear. Refer to Procedure 001-019 in Section 1. (</qs3/pubsys2/xml/en/procedures/216/216-001-019.html>)

Inspect the connecting rod bearings. Refer to Procedure 001-005 in Section 1. (</qs3/pubsys2/xml/en/procedures/220/220-001-005.html>)

Inspect the main bearings. Refer to Procedure 001-006 in Section 1. (</qs3/pubsys2/xml/en/procedures/216/216-001-006.html>)

Note : Inspect the crankshaft gear for pitting, fretting, or missing teeth. The rear crankshaft gear is not replaceable. If fretting can not be removed using Cummins® abrasive pad, Part Number 3823258, or equivalent, the crankshaft must be replaced.

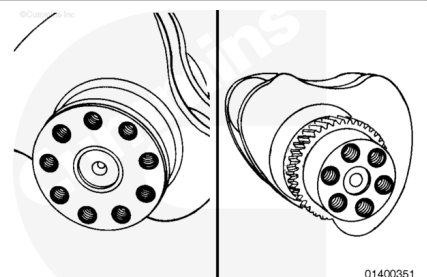


⚠ CAUTION ⚠

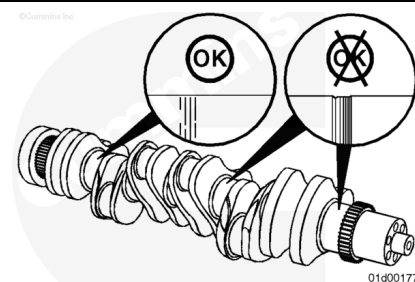
Do not chase threads on the crankshaft. Severe engine damage can occur.

Check the threads for damage at both ends of the crankshaft.

If damage is found, replace the crankshaft.



Check the main bearing journals and the rod bearing journals for damage or excessive wear. Minor scratches are acceptable.



Measure

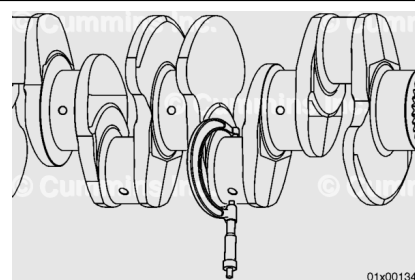
Measure the crankshaft main and connecting rod journals.

Out-of-Round: 0.050 mm [0.002 in].

Taper: 0.013 mm [0.0005 in].

Note : If crankshaft journals are **not** within the given specification, the crankshaft **must** be reground. **Always** grind all the journals when one is **not** within specifications.

Measure and record the diameter of the connecting rod journal on the crankshaft.



Standard Crankshaft Connecting Rod Journal
Diameter

	mm		in
Standard	68.987	MIN	2.7160
	69.027	MAX	2.7176

Undersize Crankshaft Connecting Rod
Journal Diameter

	mm		in	
Undersi ze	0.25 mm [0.010 in]	68.712	MIN	2.7052
	68.774	MAX	2.7076	
0.50 mm [0.020 in]	68.462	MIN	2.6954	

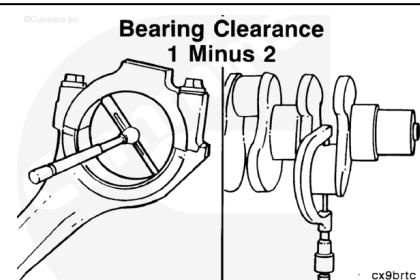
	68.527	MAX	2.6979
0.75 mm [0.030 in]	68.212	MIN	2.6855
	68.274	MAX	2.6879
1.00 mm [0.040 in]	67.962	MIN	2.6767
	68.027	MAX	2.6782

Note : If the crankshaft connecting rod journals are not within the given specification, the crankshaft must be reground. Select the target undersize journal diameter and its corresponding oversize bearing. Machine the crankshaft to the maximum journal diameter specified in the table above. Then finish the crankshaft to the target diameter. **Always** grind all the journals when one is not within specifications.

Measure and record the diameter of the connecting rod crankshaft bore with the bearings installed. The following procedure contains information for measuring the connecting rod crankshaft bore (with bearings). Refer to Procedure 001-014 in Section 1. (</qs3/pubsys2/xml/en/procedures/216/216-001-014-tr.html>)

Calculate the connecting rod bearing to crankshaft journal clearance.

Bearing clearance equals the connecting rod crankshaft bore (with bearing) minus the crankshaft connecting rod journal diameter.

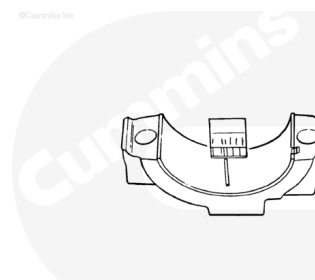


Connecting Rod to Crankshaft Bearing Clearance		
mm		in
0.04	MIN	0.002
0.12	MAX	0.005

Bearing clearance can also be determined with a plastigauge during crankshaft installation.

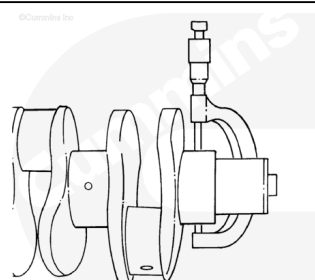
Note : If the clearance is found to be out of specification:

1. Replace or try another set of connecting rod bearings
2. Measure and inspect the connecting rod crankshaft bore with the bearings removed. Refer to Procedure 001-014 in Section 1. (</qs3/pubsys2/xml/en/procedures/220/220-001-014.html>)



cx900tc

Measure and record the diameter of the main bearing journal on the crankshaft.



ks900nc

Standard Crankshaft Main Bearing Journal Diameter

	mm		in
Standard	82.962	MIN	3.2662
	83.013	MAX	3.2682

Undersize Crankshaft Main Bearing Journal Diameter

	mm		in	
Undersize	0.25 mm [0.010 in]	82.712	MIN	3.2564
	82.763	MAX	3.2584	
0.50 mm [0.020 in]	82.462	MIN	3.2465	
	82.513	MAX	3.2485	
0.75 mm [0.030 in]	82.212	MIN	3.2367	
	82.263	MAX	3.2387	
1.0 mm [0.040 in]	81.962	MIN	3.2268	
	82.013	MAX	3.2289	

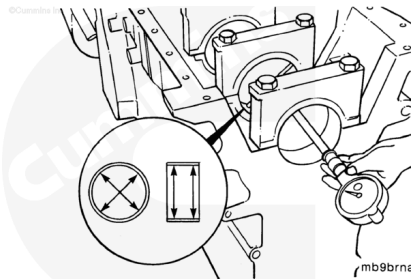
Note : If the crankshaft main bearing journals are not within the given specifications, the crankshaft must be reground. Always grind all the journals when one is not

within specifications. Oversize connecting rod bearings are available; see the appropriate parts catalog..

Install the main bearing caps with the upper and lower main bearings. Refer to Procedure 001-006 in Section 1 for main bearing cap installation.

(/qs3/pubsys2/xml/en/procedures/216/216-001-006.html)

Tighten the main bearing cap capscrews.



Torque Value: 176 n•m [130 ft-lb]

Measure the main bearing bore with the bearings installed.

Calculate the main bearing to crankshaft journal clearance.

Bearing clearance equals the main bearing crankshaft bore (with bearing) minus the crankshaft main bearing journal diameter.

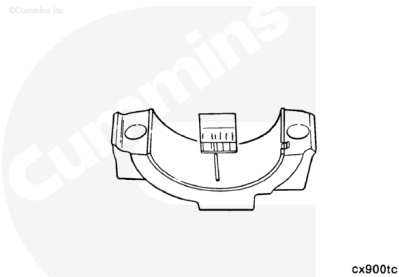
Main Bearing Bore to Crankshaft Bearing Clearance

mm		in
0.04	MIN	0.002
0.12	MAX	0.005

Bearing clearance can also be determined with a plastigauge during engine assembly.

Note : If the clearance is found to be out of specification:

1. Replace or try another set of main bearings
2. Measure and inspect the main bearing bore with the bearings removed. Refer to Procedure 001-026 in Section 1. (/qs3/pubsys2/xml/en/procedures/216/216-001-026.html)

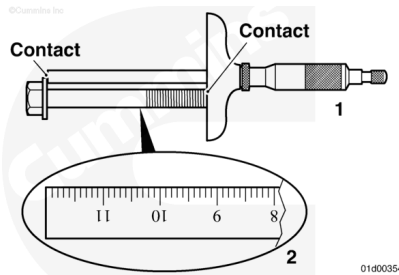


⚠ CAUTION ⚠

This step must be completed on 6.7L engines. Failure to check the main bearing capscrew against reuse guidelines can result in severe engine damage.

Main Bearing Capscrew Reuse Measurement

To check if a main bearing capscrew can be reused, the length **must** be measured by performing the following:



For each main bearing capscrew that has been removed, measure the length from underneath the head of the capscrew to the tip of the capscrew, as illustrated, using one of two methods.

1. A depth micrometer (preferred method for accuracy)
2. A machinist's rule.

If the measurement is above the maximum specification, the main bearing capscrew **must** be replaced.

Main Bearing Underhead Capscrew Length		
mm		in
120.00	MAX	4.724

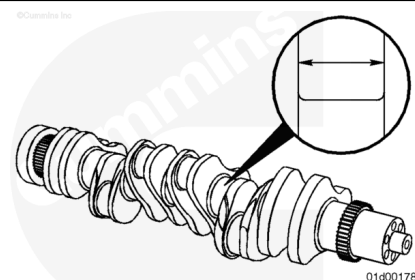
Check the thrust bearing surfaces for damage or excessive wear. Minor scratches are acceptable. Use a fine crocus cloth to polish the machined surfaces.

The thrust bearing surfaces are located at the number 6 crankshaft main bearing journal.

Measure the thrust distance.

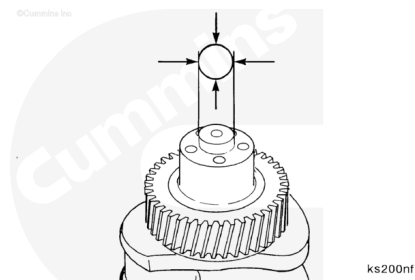
Crankshaft Thrust Face Width (Standard)		
mm		in
37.48	MIN	1.475
37.55	MAX	1.478

Note : Oversize thrust bearings are available if the thrust distance is not within specifications. See the appropriate parts catalog.



Measure the damper pilot outside diameter.

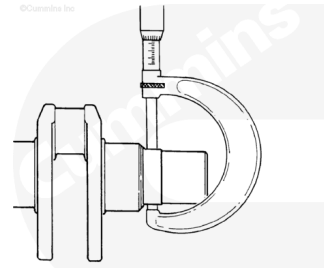
Crankshaft Damper Pilot Outside Diameter		
mm		in
24.92	MIN	0.981
25.00	MAX	0.984



Measure the crankshaft gear journal outside diameter.

Crankshaft Gear Journal Outside Diameter

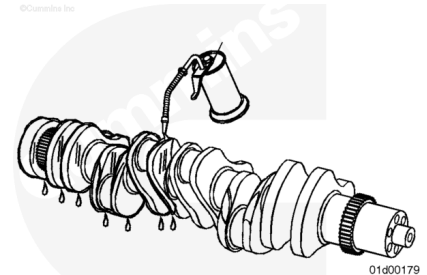
mm		in
70.59	MIN	2.779
70.61	MAX	2.780



ks1getb

Use a light preservative oil to lubricate the crankshaft to prevent rust.

Note : If the crankshaft is **not** going to be used immediately, use a **heavy** preservative oil. Protect the crankshaft with a cover to prevent dirt from sticking to the oil.

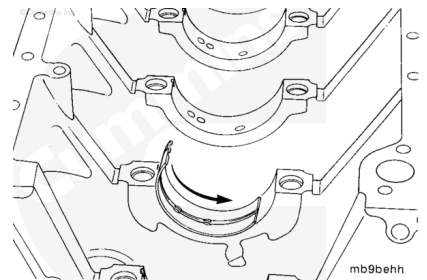


01d00179

Install

⚠ CAUTION ⚠

The tang (1) on the bearing shell must be in the slot (2) of the bearing saddle to correctly position the bearing and prevent engine damage.



mb9behh

Upper Main Bearings

Make sure the backside of the bearings are clean and free of debris before installing the upper main bearings into the block.

Do **not** lubricate the side of the main bearing that is against the cylinder block.

Apply a coat of assembly lubricant, Part Number 3163087, or equivalent, to the crankshaft side of the upper main bearings.

Note : Make sure the main bearing being installed is the same size as the main bearing that was removed. The size is engraved on the back of the main bearing.

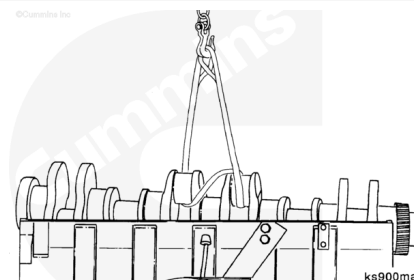
Note : The crankshaft thrust bearing must be installed in the number 6 position.

Note : The upper and lower main bearing shells of some engines are **not** interchangeable. The backs of the main bearings are marked with the proper orientation, if required.

Note : If used bearing shells are to be installed, they must be installed in their original locations, as marked during disassembly.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ CAUTION ⚠

Carefully install the crankshaft to avoid damage to the crankshaft main bearings, especially the thrust/main bearing journals. Engine life will be shortened if damage to the crankshaft occurs.

Use a hoist and nylon list sling, Part Number 3375957.

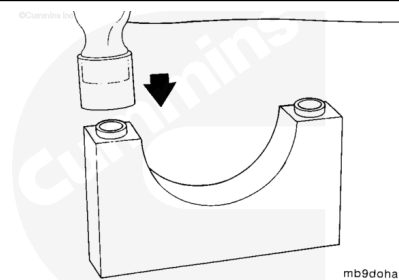
Install the sling around the number 3 and number 4 connecting rod bearing journals.

Install the crankshaft.

Note : Lower the crankshaft straight down to avoid damage to the crankshaft and cylinder block.

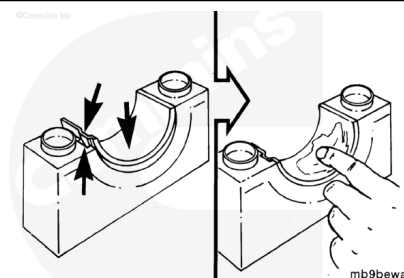
Note : When setting the crankshaft into the block, make sure the front crankshaft gear meshes with the lubricating oil pump gear, if not previously removed.

Check the main bearing caps to make sure the ring dowels are installed.



⚠ CAUTION ⚠

Do not lubricate the back side of the bearing that contacts the main bearing cap.



Make sure the backside of the bearings are clean and free of debris before installing the lower main bearings into the main bearing caps.

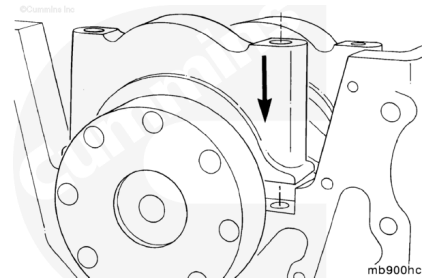
Make sure to align the tang of the bearings with the recesses on the main bearing caps.

Note : Some engines use a thrust bearing for the upper and lower main bearing (360 degree), while other engines only have a thrust bearing for the upper main bearing (180 degree). Always install replacement bearings of the same design.

If equipped, install the lower crankshaft thrust bearing at the number 6 main bearing position.

Apply a coat of Cummins® assembly lube, Part Number 3163087, to the crankshaft side of the main bearings and thrust bearing surfaces.

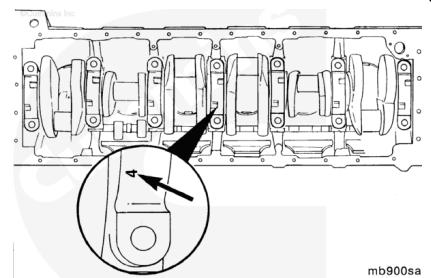
Make sure the main bearing cap surfaces between the main bearing cap and block are clean and free of debris.



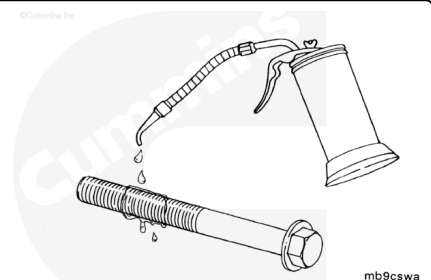
The main bearing caps are/were numbered during the removal process for their location. Number 1 starts with the front of the block.

Note : The caps must be installed so the numbers on the caps match the bearing saddle in the block. The lock tangs in the main bearing saddle and bearing cap must be on the same side.

Install the main bearing caps. Make sure to align the ring dowels on the main bearing cap with the corresponding drillings in the cylinder block.



Lubricate the main bearing capscrew threads and underside of the head with clean engine oil.

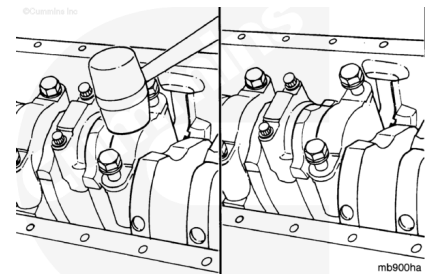


Gently tap the main bearing cap into position with a plastic or rubber mallet.

When seated, install the main bearing capscrews and hand-tighten.

Torque Value: 50 n•m [37 ft-lb]

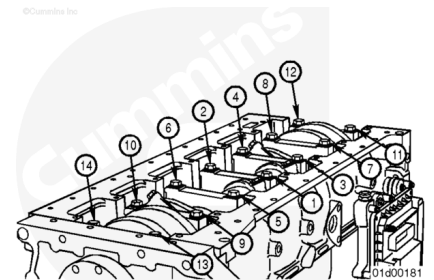
Do **not** tighten to the final torque value at this time. Final torque **must** be applied after all main bearing caps are installed.



Note : The sequence to the right is for a 6 cylinder engine. For a 4 cylinder engine, use the same sequence for the 5 main bearing caps.

⚠ CAUTION ⚠

There is a different torque procedure for new and previously installed main bearing capscrews. Failure to use the correct torque value can result in engine damage.



Tighten the capscrews evenly and in sequence. Perform each step to all capscrews before performing the next step. As illustrated.

Torque Value:

Previously Installed Main Bearing Capscrews

1. 50 mm [37 in]
2. 60 n•m [44 ft-lb]
3. 80 n•m [59 ft-lb]
4. Rotate 90 degrees.

Torque Value:

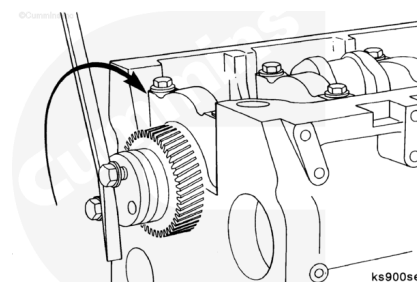
New Main Bearing Capscrews

1. 50 mm [37 in]
2. 60 n•m [44 ft-lb]
3. 80 n•m [59 ft-lb]
4. Rotate 90 degrees.
5. 120 n•m [89 ft-lb]
6. Loosen completely.
7. 60 n•m [44 ft-lb]
8. 85 n•m [63 ft-lb]
9. Rotate 120 degrees.

The crankshaft **must** rotate freely after installing the main bearing caps.

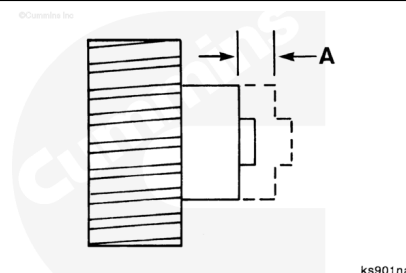
If the crankshaft does **not** rotate freely:

1. Check if the crankshaft is contacting one of the connecting rods
2. Check if the correct main bearing caps were installed correctly
3. Check if the main bearing cap ring dowels or mounting surfaces were damaged during installation
4. Check if the correct main bearings were installed.



Note : The dimensions of the thrust bearing and crankshaft journal determine end clearance.

Measure the crankshaft end clearance with a Cummins® dial indicator assembly, Part Number 3824564, and magnetic base, Part Number 3377399.



Crankshaft End Clearance

mm		in
0.065	MIN	0.0026
0.415	MAX	0.0163

If the crankshaft end clearance **not** within specification:

1. If the crankshaft end clearance is below specification, check if there are any obstructions limiting the crankshaft's travel (lubricating oil pump, connecting rod, etc.)
2. If the crankshaft end clearance is above specification, inspect the crankshaft thrust bearing surface. Also check if the correct thrust bearing(s) were installed.

Note : Oversize thrust bearings are available if the end clearance **not** within specifications. Oversize thrust bearings of 0.25 to 0.51 mm [0.010 to 0.020 in] are available.

Finishing Steps

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged

contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

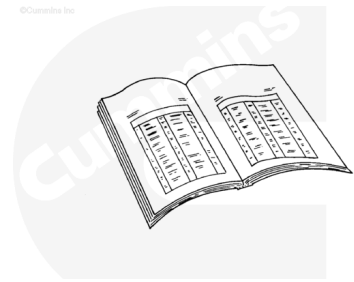
⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Install the rear gear housing. Refer to Procedure 001-034 in Section 1. (</qs3/pubsys2/xml/en/procedures/216/216-001-034-tr.html>)
- Install the connecting rod caps. Refer to Procedure 001-005 in Section 1. (</qs3/pubsys2/xml/en/procedures/220/220-001-005.html>)
- Install the camshaft gear. Refer to Procedure 001-012 in Section 1. (</qs3/pubsys2/xml/en/procedures/216/216-001-012-tr.html>)
- Install the fuel pump and gear. Refer to Procedure 005-016 in Section 5. (</qs3/pubsys2/xml/en/procedures/216/216-005-016-tr.html>)
- If equipped, install or connect any driven accessories (i.e., hydraulic pump).
- Install the lubricating oil suction tube. Refer to Procedure 007-035 in Section 7. (</qs3/pubsys2/xml/en/procedures/220/220-007-035-tr.html>)
- Install the block stiffener plate. Refer to Procedure 001-089 in Section 1. (</qs3/pubsys2/xml/en/procedures/220/220-001-089.html>)
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/216/216-007-025-tr.html>)



ck800wa

- For front gear train engines, install the rear seal and rear seal carrier. Refer to Procedure 001-067 in Section 1. (</qs3/pubsys2/xml/en/procedures/220/220-001-067.html>)
- Install the flywheel housing. Refer to Procedure 016-006 in Section 16. (</qs3/pubsys2/xml/en/procedures/216/216-016-006-tr.html>)
- Install the rear seal. Refer to Procedure 001-024 in Section 1. (</qs3/pubsys2/xml/en/procedures/216/216-001-024-tr.html>)
- Install the flywheel or flexplate. Remove the flywheel or flexplate. Refer to Procedure 016-005 (</qs3/pubsys2/xml/en/procedures/216/216-016-005-tr.html>) or Refer to Procedure 016-004 in Section 16. (</qs3/pubsys2/xml/en/procedures/216/216-016-004-tr.html>)
- Install the starter. Refer to Procedure 013-020 in Section 13. (</qs3/pubsys2/xml/en/procedures/216/216-013-020-tr.html>)
- Install the front cover. Refer to Procedure 001-031 in Section 1. (</qs3/pubsys2/xml/en/procedures/216/216-001-031-tr.html>)
- Install the vibration damper. Refer to Procedure 001-052 in Section 1. (</qs3/pubsys2/xml/en/procedures/216/216-001-052-tr.html>)
- If required, install the fan hub pulley. Refer to Procedure 008-039 in Section 8. (</qs3/pubsys2/xml/en/procedures/216/216-008-039-tr.html>)
- Check the engine rotation. See the Rotation Check section of this procedure located below.
- Install the drive belt. Refer to Procedure 008-002 in Section 8. (</qs3/pubsys2/xml/en/procedures/220/220-008-002-tr.html>)
- Install the engine. Refer to Procedure 000-002 in Section 0. (</qs3/pubsys2/xml/en/procedures/220/220-000-002.html>)
- Fill the Lubricating oil pan with oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/216/216-007-037.html>)
- Connect the batteries. Refer to the OEM service manual.
- Check for leaks and proper engine operation.

Rotation Check

With the engine fully assembled, check that the engine rotates freely. Use a Cummins® barring tool, Part Number 3824591.

Insert the barring tool into the flywheel housing and engage the flywheel/flexplate ring gear. The crankshaft can then be rotated by hand. Use a 1/2- inch drive ratchet or breaker bar.

If the engine does **not** rotate freely, check for any external obstructions. (Flywheel/flexplate, engine driven accessories, etc.) If no obstructions are found, remove the oil pan and look for internal damage.

